

Auteur: Michael Livchits
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$$I = \int_1^{+\infty} \frac{\ln x}{x^2} dx$$

Eerst vinden we de onbepaalde integraal

$$\int \frac{\ln x}{x^2} dx$$

Partiele integratie:

$$\begin{aligned} \begin{cases} u = \ln x \\ dv = \frac{1}{x^2} dx \end{cases} &\Rightarrow \begin{cases} du = \frac{1}{x} dx \\ v = -\frac{1}{x} \end{cases} \\ \int \frac{\ln x}{x^2} dx &= -\frac{\ln x}{x} + \int \frac{dx}{x^2} = -\frac{\ln x}{x} - \frac{1}{x} \end{aligned}$$

Neem de limiet

$$\begin{aligned} I &= \lim_{b \rightarrow +\infty} \left[-\frac{\ln x}{x} - \frac{1}{x} \right]_1^b \\ &= \lim_{b \rightarrow +\infty} \left(-\frac{\ln b}{b} - \frac{1}{b} + 1 \right) \\ &= \lim_{b \rightarrow +\infty} -\frac{\ln b}{b} + \lim_{b \rightarrow +\infty} \left(-\frac{1}{b} + 1 \right) \end{aligned}$$

l'hopital toepassen op eerste limiet:

$$\begin{aligned} &= \lim_{b \rightarrow +\infty} -\frac{1}{b} + \lim_{b \rightarrow +\infty} \left(-\frac{1}{b} + 1 \right) = 1 \\ &\Rightarrow \text{convergeert} \end{aligned}$$

References